

Computationally Efficient Predictions for Multi-Output Spatiotemporal Dynamical Systems

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September 30, 2026

Abstract

The abstract is here, which should have no citations/reference.

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1 Introduction

The paper *Efficient spatio-temporal Gaussian regression via Kalman filtering* (Todescato et al., 2020) introduces a methodology for estimating future states of a spatio-temporal dynamical system using a combination of Gaussian process regression and Kalman filtering to account for spatial and temporal aspects which guide the ‘evolution’ of the dynamical system of interest.

2 Background

2.1 Gaussian Processes

A Gaussian process (GP) is a real-valued stochastic process indexed over some D -dimensional *state space* $\mathcal{X} \subseteq \mathbb{R}^D$ which satisfies the following key property:

Definition 2.1. *For any finite collection (N -tuple) of states $(x_1, \dots, x_N) \subset \mathcal{X}$, the joint distribution of the vector $(f(x_1), \dots, f(x_N))$ follows a multivariate Gaussian distribution on $\mathbb{R}^N$.*

Furthermore, every Gaussian process on the state space $\mathcal{X}$ is fully specified by two real-valued functions on $\mathcal{X}$, which respectively describe the mean and covariance of the underlying process (Rasmussen and Williams (2006)). Formally, every Gaussian process defined on the underlying state space $\mathcal{X}$ is *uniquely* described by a mean function $\mu : \mathcal{X} \rightarrow \mathbb{R}$ and a covariance (kernel) function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ which provide the following structure:

- **Mean:** $\mu(x) = \mathbb{E}[Y(x)]$ for all $x \in \mathcal{X}$.
- **Covariance:** $K(x, x') = \text{Cov}(Y(x), Y(x'))$ for all $(x, x') \in \mathcal{X} \times \mathcal{X}$.

As every Gaussian process on $\mathcal{X}$ is uniquely determined by these two functions, we may simply write that $f \sim \mathcal{GP}(\mu, K)$ to denote a Gaussian process with mean μ and kernel K .

Due to these convenient properties, Gaussian processes are frequently used in a variety of practical statistical applications, ranging from geostatistics and climatology to fields such as electrical engineering and signal processing (Särkkä and Hartikainen (2012); Huang et al. (2025)).

Gaussian process regression, often referred to in the literature as *kriging* after the South African geostatistician Danie G. Krige (Cressie, 1990; Guhaniyogi and Banerjee, 2018), is a semi-parametric statistical modelling procedure wherein a finite collection of observations are used for learning the outputs of an unknown function defined on the state space $\mathcal{X}$.

2.2 Kalman Filtering

The Kalman filter (KF), first introduced by and eponymously named for Rudolf E. Kálmán (1960), is a technique for

3 Problem Formulation

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4.1 Replication

4.2 Extension

5 Results

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6.1 Limitations

7 Discussion

7.1 Future Work

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Appendix

A Definitions and Technical Details

B Code and Simulation Outputs

For the sake of brevity, the code used in this project is not directly added to the appendix. All code is available in a public GitHub repository, which can be located at <https://github.com/hotz1/Dynamical-Multitask-Prediction>.